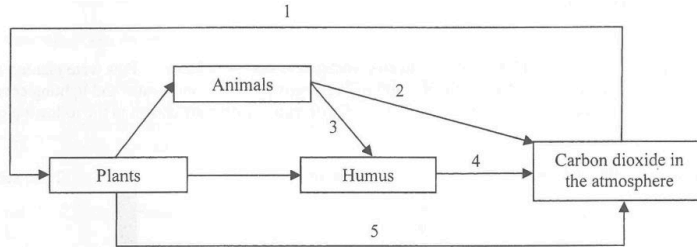


1. 2012Q16

Questions 16 and 17 refer to the following diagram, which shows the flow of carbon in an ecosystem:



Which of the following combinations correctly shows the processes indicated by labels 1, 4 and 5?

- | | | | |
|----|----------------|----------------|---------------|
| | 1 | 4 | 5 |
| A. | respiration | photosynthesis | decomposition |
| B. | photosynthesis | respiration | decomposition |
| C. | photosynthesis | decomposition | respiration |
| D. | decomposition | photosynthesis | respiration |

2. 2012Q17

Saprophytes are involved in process

- A. 1.
B. 2.
C. 3.
D. 5.

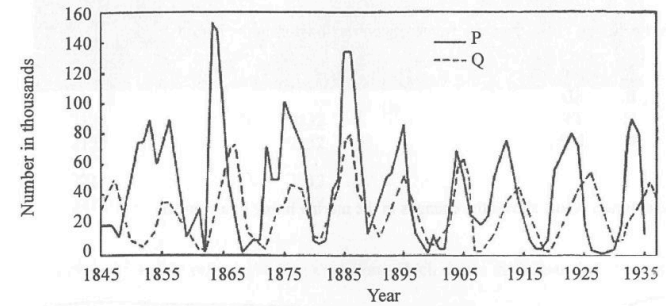
3. 2014Q30

Some environmental protection groups claim that the vegetarian diet is good for our environment. This is probably because:

- A. vegetables grow faster than animals.
B. it protects endangered species as less animals are killed for food.
C. growing vegetables can produce oxygen but rearing animals only consumes oxygen.
D. it reduces the emission of carbon dioxide associated with rearing animals for food.

4. 2014Q31

The graph below shows the changes in the populations of two organisms that exhibit a predator-prey relationship in a habitat:



Which of the following statements about the identity of the organism is correct?

- A. P is the predator because its number fluctuates more than that of Q.
B. P is the predator because its lowest number is lower than that of Q.
C. Q is the predator because its number fluctuates less than that of P.
D. Q is the predator because its highest number is lower than that of P.

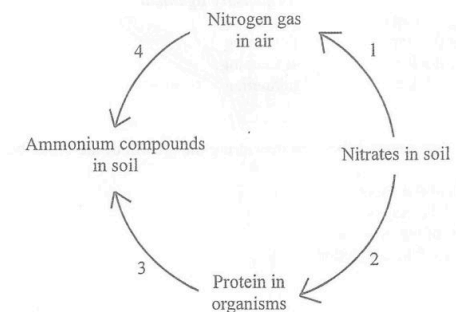
5. 2014Q33

Which of the following processes releases nitrogen-containing compounds from organisms back into the environment?

- A. Nitrification
B. Decomposition
C. Denitrification
D. Nitrogen fixation

6. 2015Q30

Questions 30 and 31 refer to the diagram below, which shows the conversion of some nitrogen-containing substances in nature:



Process 3 is

- A. nitrification.
B. denitrification.
C. decomposition.
D. nitrogen fixation.

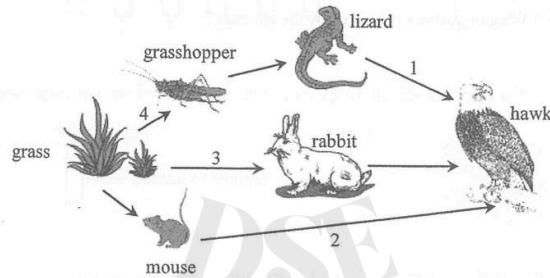
7. 2015Q31

Which of the following processes does **not** require the action of bacteria?

- A. 1
- B. 2
- C. 3
- D. 4

8. 2016Q30

The following is a food web found in grassland:



If the energy input for each food chain is the same, the transfer of energy between two trophic levels is the smallest in

- A. 1.
- B. 2.
- C. 3.
- D. 4.

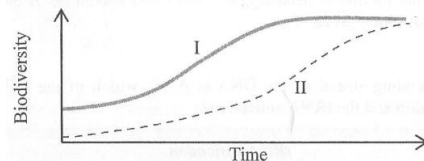
9. 2016Q34

Symbiotic bacteria found in the root nodules of bean plants are able to convert:

- A. nitrite ions to nitrate ions.
- B. nitrogen gas to nitrate ions.
- C. organic nitrogenous compounds to nitrate ions.
- D. nitrogen gas to ammonium ions.

10. 2017Q12

The following graph shows the predicted change in biodiversity during the process of primary and secondary succession in an ecosystem:

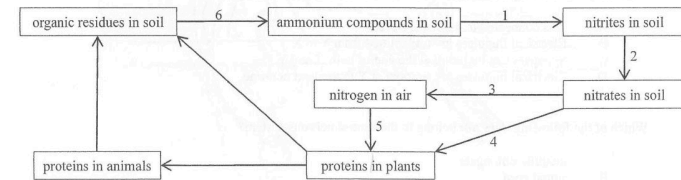


Which of the following combinations is correct?

	Type of succession	Explanation
A.	I is primary succession.	A climax community is reached in I.
B.	I is secondary succession.	I has a higher biodiversity than II at the beginning.
C.	II is primary succession.	Pioneer community is involved in II.
D.	II is secondary succession.	II shows a slow increase in biodiversity.

11. 2017Q32

Questions 32 and 33 refer to the diagram below, which shows some processes in the nitrogen cycle in nature:



Which of the following combinations correctly matches the microorganism involved in processes 3 and 6?

	3	6
A.	denitrifying bacteria	nitrogen-fixing bacteria
B.	Nitrogen-fixing bacteria	fungi
C.	fungi	denitrifying bacteria
D.	denitrifying bacteria	fungi

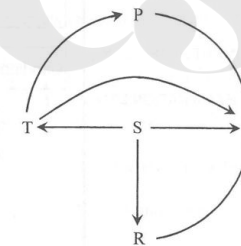
12. 2017Q33

Which of the following processes **does not** require the action of bacteria?

- A. 1
- B. 2
- C. 4
- D. 5

13. 2017Q34

Questions 34 and 35 refer to the diagram below, which shows the flow of energy among organisms found in a grassland:



Which of the following statements correctly describe(s) the above organisms?

- (1) S is a heterotroph.
 - (2) P is the secondary consumer.
 - (3) Q has the largest total biomass.
- A. (1) only
 - B. (2) only
 - C. (1) and (3) only
 - D. (2) and (3) only

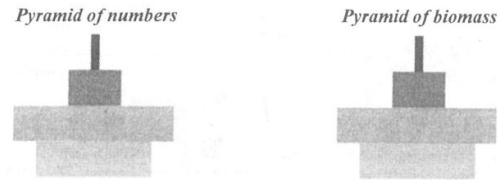
14. 2017Q35

If there is a decomposer in this group of organisms, it is most likely to be

- A. Q.
- B. R.
- C. S.
- D. T.

15. 2018Q5

The diagrams below show the pyramids of numbers and biomass of a food chain:



Which of the following is most likely to be the producer in this food chain?

- A. rice
- B. trees
- C. grass
- D. phytoplankton

16. 2018Q6

Protecting sharks in the wild is important for maintaining the ecological balance of the marine ecosystem because

- A. sharks are an endangered species.
- B. shark fin is a popular dish for banquets.
- C. the dead bodies of sharks are an important food source for decomposers.
- D. sharks are top predators that regulate the population sizes of other consumers.

17. 2018Q14

Caterpillar fungus is a kind of Chinese herbal medicine. When the spores of this fungus land on the larvae of moths, the spores will germinate inside the body of the larvae and grow out of their heads, as shown in the photograph below:



Which of the following best describes the role of this fungus?

- A. parasite
- B. predator
- C. producer
- D. consumer

18. 2018Q32

Organisms P and Q are found in the same local habitat. Their population sizes have continued to grow in the last few years. Which of the following statements best describes organisms P and Q?

- A. They are heterotrophs.
- B. They are top predators.
- C. They have different niches.
- D. They have different predators.

19. 2018Q33

Which of the following statements about primary succession and secondary succession is correct?

- A. Primary succession is always followed by secondary succession.
- B. Primary succession always starts with a barren area while secondary succession does not.
- C. Secondary succession always ends with a climax community while primary succession does not.
- D. Secondary succession always ends with a climax community while primary succession ends with a pioneer community.

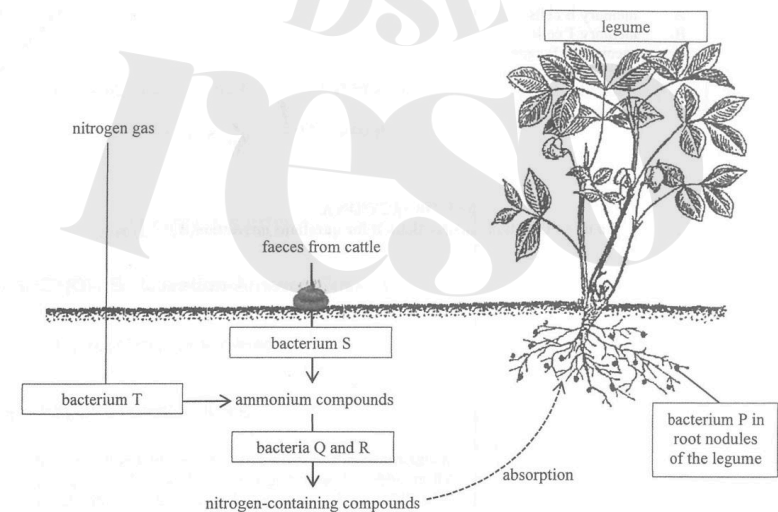
20. 2019Q29

If a student wants to find out whether the distribution of plant species is affected by the slope of a hillside, which of the following sampling methods should be used?

- A. Set up a line transect along the hillside and record the plant species that touch the line.
- B. Set up a line transect around the hillside and record the plant species that touch the line.
- C. Randomly place quadrats along the hillside and record the plant species within the quadrat.
- D. Randomly place quadrats around the hillside and record the plant species within the quadrat.

21. 2019Q33

Questions 33 and 34 refer to the diagram below, which shows some processes of the nitrogen cycle:



Which of the following statements about the bacteria shown in the diagram is correct?

- A. P belongs to parasitic bacteria.
- B. Q belongs to nitrifying bacteria.
- C. R belongs to nitrogen-fixing bacteria.
- D. S belongs to denitrifying bacteria.

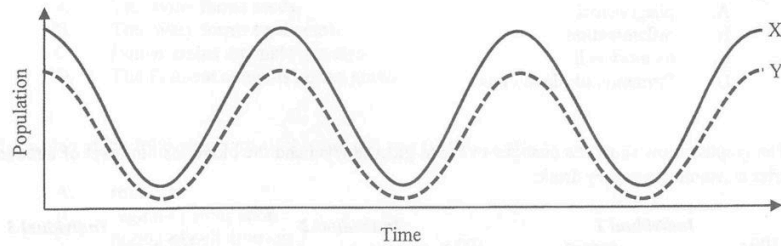
22. 2019Q34

Which of the following bacteria has a similar role to fungi in the cycling of materials?

- A. Q
- B. R
- C. S
- D. T

23. 2020Q32

The following graph shows the change in the populations of organisms X and Y living in the same habitat:



Which of the following is most likely the ecological relationship between organisms X and Y?

- A. predation
- B. mutualism
- C. competition
- D. commensalism

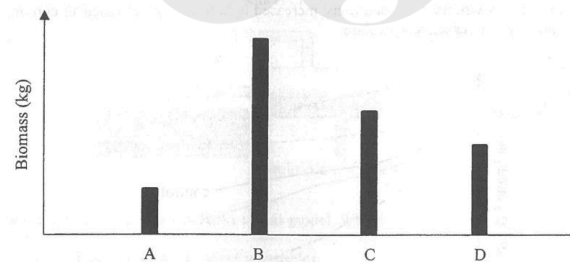
24. 2021Q32

During the primary succession in a terrestrial habitat, which of the following descriptions about the ecosystem is **incorrect**?

- A. The nutrient level of the soil decreases.
- B. The complexity of the food web increases.
- C. The abundance of non-vascular plants decreases.
- D. The energy available to the community increases.

25. 2021Q33

Questions 33 and 34 refer to the following graph, which shows the biomasses of four populations forming a food chain in a terrestrial habitat.



Which population is most likely the autotrophs in this habitat?

- A. A
- B. B
- C. C
- D. D

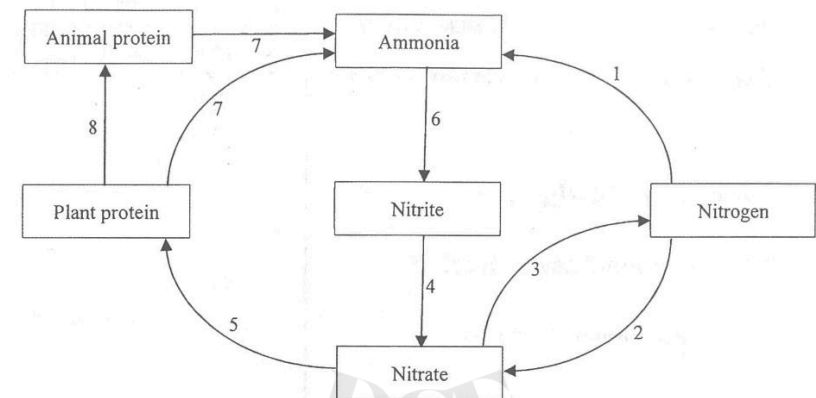
26. 2021Q34

What would you expect to happen if the number of individuals in population D decreased?

- A. The number of individuals in population A would increase.
- B. The number of individuals in population C would decrease.
- C. The amount of energy passed to population A would decrease.
- D. The amount of energy passed to population C would increase.

27. 2021Q35

Questions 35 and 36 refer to the diagram below, which shows some natural processes involved in the nitrogen cycle:



Process 2 refers to

- A. lighting.
- B. nitrification.
- C. denitrification.
- D. nitrogen fixation.

28. 2021Q36

Which of the following pairs of processes involves the action of bacteria?

- A. 1 and 5
- B. 2 and 6
- C. 3 and 7
- D. 4 and 8

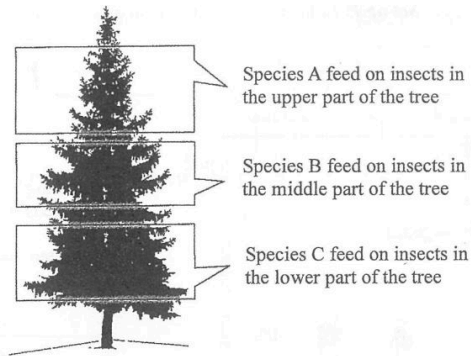
29. 2022Q8

Decomposers can speed up the process of ecological succession because they:

- A. break down rocks into soil particles.
- B. remove dead organic matter from the soil.
- C. release carbon back into the atmosphere in the form of carbon dioxide.
- D. release nitrogen back into the soil in the form of ammonium compounds.

30. 2022Q9

Three bird species are living on the same tree species for years. They feed on the insects found at different regions of the tree as shown in the diagram below:

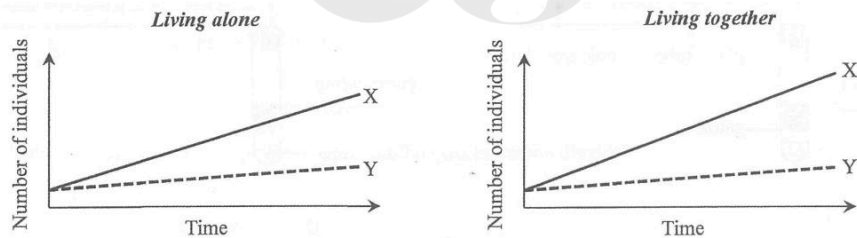


Which of the following can be deduced from the above phenomenon?

- (1) The insects are evenly distributed on the tree.
 - (2) Birds species A, B and C have different ecological niches.
 - (3) There is less interspecific competition among bird species A, B and C.
- A. (1) and (2) only
 - B. (1) and (3) only
 - C. (2) and (3) only
 - D. (1), (2) and (3)

31. 2022Q10

The graph below show the changes in the population sizes of organisms X and Y over a period of time when they are living alone or living together:

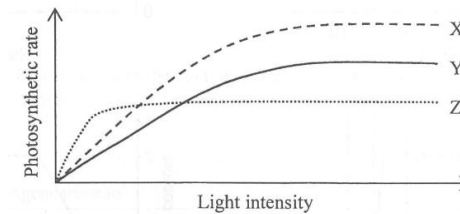


Which of the following correctly describes the ecological relationship of X and Y?

- A. predation
- B. mutualism
- C. competition
- D. commensalism

32. 2023Q10

The graph below shows the photosynthetic rates of three plant species X, Y and Z in response to the change of light intensity. The three plant species are in the same habitat.



Which of the following can be deduced from the above graph?

- A. Species Y is the tallest among the three species.
- B. Species Z is most competitive in shady condition.
- C. Species Y will outcompete species X in all light conditions.
- D. Removal of species Z will increase the photosynthetic rate of species X.

33. 2023Q21

Kitchen wastes can be converted to fertilizers. Which of the following processes are essential for this conversion?

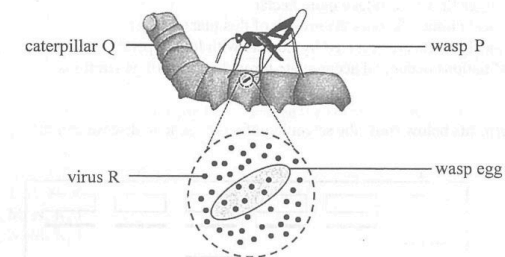
- A. nitrification and decomposition
- B. nitrification and nitrogen fixation
- C. denitrification and decomposition
- D. denitrification and nitrogen fixation

34. 2023Q22

Questions 22 and 23 refer to the following information about the interactions of wasp P, caterpillar Q, and virus R.

Wasp P lays eggs inside the caterpillar Q. When the eggs hatch, the wasp larvae feed on the body tissues of the caterpillars and kill them eventually.

Recent research showed that these wasps are the host of virus R. The viral DNA becomes part of the wasp genome and passes to the wasp's offspring. The wasps inject their eggs along with the virus R into the caterpillars. Virus R protect the eggs by suppressing the caterpillars' body defense mechanism.



Which of the following correctly describes their feeding roles?

- A. The virus is the top consumer.
- B. The caterpillar is the producer.
- C. The wasp adult is the primary consumer.
- D. The wasp larva is the secondary consumer.

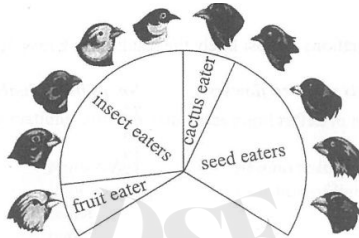
35. 2023Q23

Which of the following correctly shows the relationship between the wasp and the virus?

- A. parasitism
- B. mutualism
- C. competition
- D. commensalism

36. 2024Q14

Darwin's finches are a group of small birds isolated on the Galapagos Islands. These birds show variations in forms and function of their beak as shown in the diagram below:

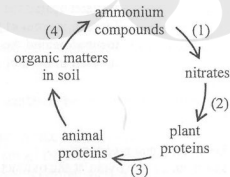


What is the significance of these variations to the survival of Darwin's finches on the Galapagos Islands?

- A. The variations promote natural selection.
- B. They can distinguish each other by their beaks.
- C. They interbreed to produce offspring with more variations.
- D. They have different ecological niches to reduce competition.

37. 2024Q32

Questions 32 and 33 refer to the diagram below, which shows the conversion of some nitrogenous compounds in nature:



Which of the following statements about the conversions is correct?

- A. Process (1) involves nitrogen fixing bacteria.
- B. Process (2) is anabolic in nature.
- C. Process (3) is catabolic in nature.
- D. Process (4) involves denitrifying bacteria.

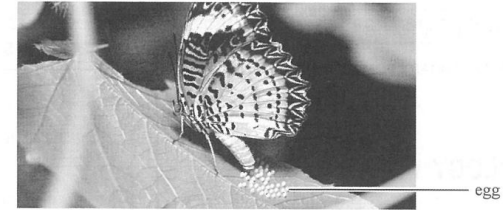
38. 2024Q33

Which of the following process(es) involve(s) assimilation?

- A. (1) only
- B. (3) only
- C. (1) and (2) only
- D. (2) and (3) only

39. 2024Q34

Question 34 refers to some information about the interaction among three organisms: butterflies, plant species X and ants. Butterflies lay eggs on the leaf surfaces of plant species X. When caterpillars hatch from the eggs, they feed on the leaves. Plant species X secretes a sugary solution on their leaves. Ants are attracted to feed on the sugary solution and the eggs.



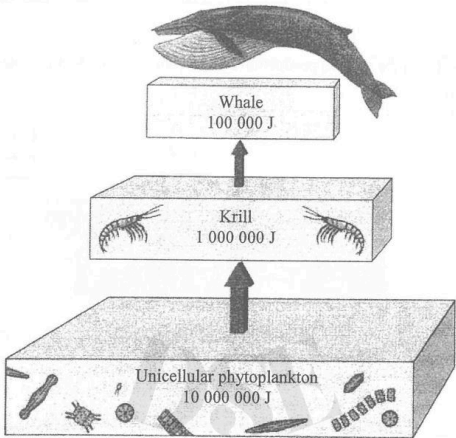
Which of the following combinations correctly describes the relationship between the organisms mentioned?

- | | |
|--------------------------------|--------------------------|
| Plant species X and ant | Ant and butterfly |
| A. mutualism | competition |
| B. mutualism | predation |
| C. commensalism | competition |
| D. commensalism | predation |

LQ

1. 2012Q6

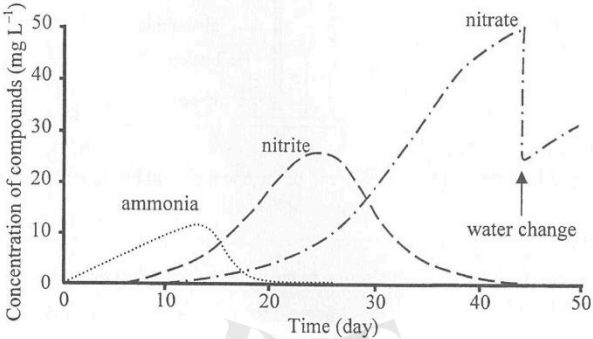
The diagram below shows a food chain in an ocean. The energy content of each trophic level is shown in the boxes:



- a) In the space below, calculate the percentage decrease in energy content from unicellular phytoplankton to krill. (2 marks)
- b) Give *two* reasons why there is a decrease in energy content from a lower trophic level to a higher one. (2 marks)
- c) In another food chain, the producer is tree and the primary consumer is caterpillar. Would the percentage decrease in energy content be greater or smaller than the value found in (a)? Explain your answer. (2 marks)

2. 2013Q5

Knowing that the nitrogen cycle is important to the healthy growth of fish, Tom purchased testing kits to monitor the levels of ammonia, nitrite and nitrate in his aquarium. He kept the aquarium aerated with an air pump. The graph below shows the changes in the concentrations of the three compounds over 50 days:



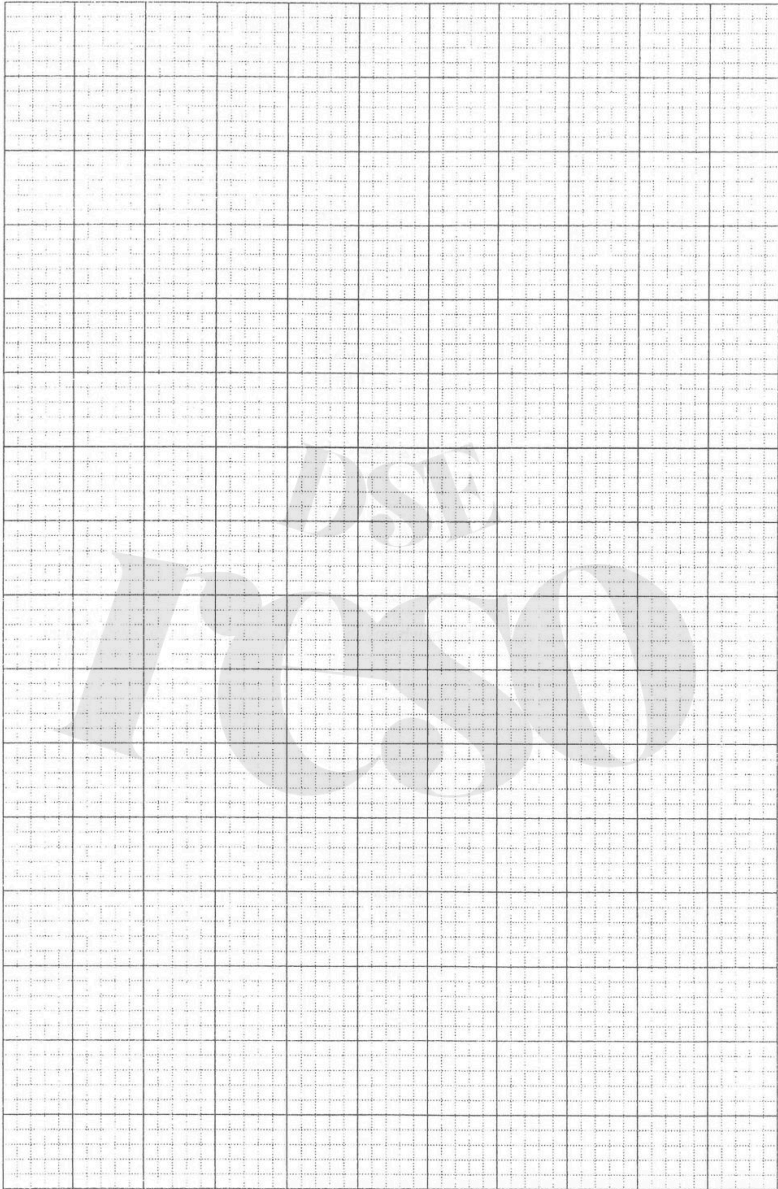
- a) Name the process that led to the changes shown in the graph. (1 mark)
- b) After three weeks, Tom observed a high concentration of nitrite in the water and the fish were showing symptoms of nitrite poisoning. Therefore, Tom added a suspension of a bacterial culture to the water to lower the nitrite concentration. Suggest the type of bacteria he added and explain how it lowered the nitrite concentration. (2 marks)
- c) To prevent the accumulation of nitrate, Tom regularly replaced some water in the aquarium with fresh water. What else could Tom do to lower the nitrate concentration? (2 marks)

3. 2013Q8

A study was carried out to investigate the distribution and abundance of animal species A and B on a rocky shore. The table below shows the results obtained from the back of the shore to the waterfront:

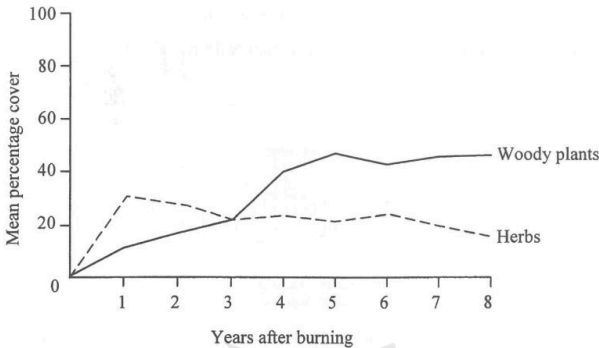
Distance from the back of the shore (m)	Abundance (number of individual m ²)	
	Species A	Species B
1	10	0
2	25	2
3	40	8
4	38	10
5	20	20
6	18	35
7	15	33
8 (waterfront)	11	28

- a) Using the graph paper on the opposite page, plot a graph to show the results. (5 marks)
- b) Based on the information in the graph, suggest which species would have a higher tolerance to desiccation. Explain your answer. (4 marks)
- c) Briefly describe the sampling procedure used to obtain the above data. (3 marks)



4. 2014Q5

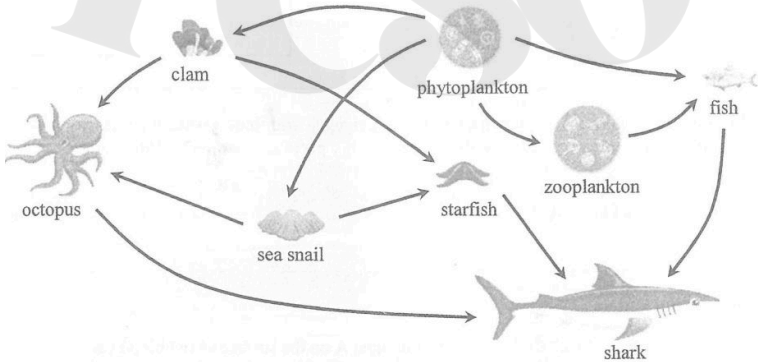
In a study, a plot of land was burnt by fire. After this, the vegetation on this land, classified into herbs and woody plants, was monitored for 8 years. The percentage cover of each type of vegetation is shown in the graph below:



- a) Which type of succession is shown in the above case? Explain your answer. (2 marks)
- b) i) Describe briefly how the dominant community of vegetation changes with time after the fire. (2 marks)
- ii) Explain the changes in the dominant community described in (i). (4 marks)

5. 2015Q3

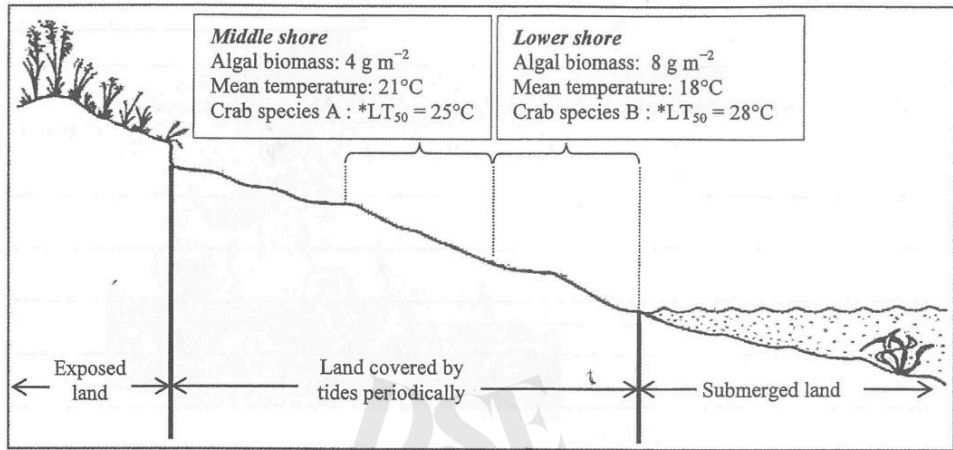
The diagram below shows the feeding relationships among some organisms in a marine ecosystem:



- a) Write down the shortest food chain found in the diagram. (1 mark)
- b) In the space provided below, draw the pyramid of numbers for the food chain in (a). (2 mark)
- c) Explain the shape of the pyramid of numbers drawn in (b). (3 mark)
- d) Suggest *two* practical methods that allow you to confirm the feeding relationships among various organisms in this ecosystem. (2 mark)

6. 2016Q5

The diagram below shows some biotic and abiotic factors of a rocky shore, the distribution of two crab species and their temperature tolerance:

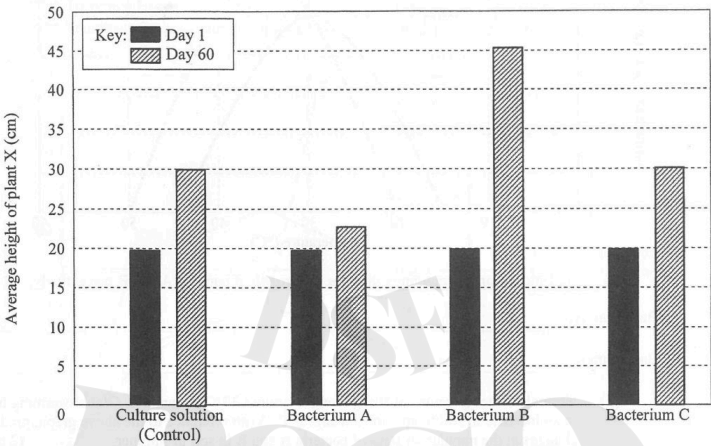


*Median lethal temperature (LT₅₀): When a species is exposed to that temperature for 24 hours, half of the individuals die.

- a) The two species coexist on a rocky shore and feed on the same species of alga. When the two crab species are kept in a simulated habitat with the algal species, they will fight against each other. According to the information given in the diagram, deduce which crab species, A or B, would be a stronger competitor. (3 marks)
- b) Deduce whether temperature tolerance is a determining factor for the distribution of these crab species. (4 marks)
- c) Explain why quadrats are unsuitable for studying the abundance of crabs on the rocky shore. (2 marks)

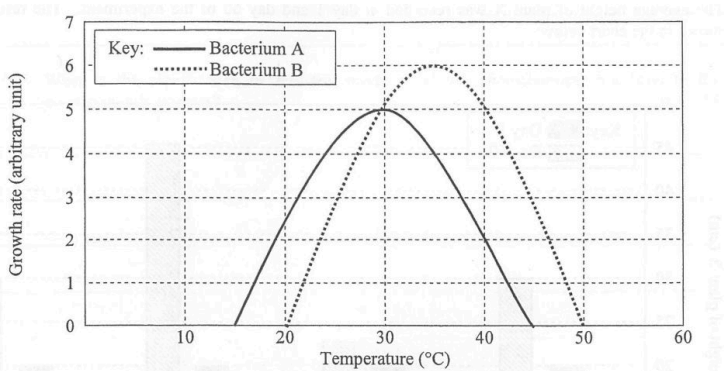
7. 2017Q8

In an investigation about the effect of soil bacteria on the growth of plant X, three types of soil bacteria (A, B, and C) were grown separately in one type of culture solution. After this, each bacterial culture was added to the soil of separate pots of plant X. A control was prepared by adding the culture solution only. The average height of plant X was recorded at day 1 and day 60 of the experiment. The results are shown in the chart below:



- a) With reference to the above results, state the effect of each type of bacterium on the growth of plant X. (3 marks)
Bacterium A: Bacterium B: Bacterium C:
- b) It is known that both bacteria A and B are able to colonise the root of plant X and obtain nutrients from the root for growth. Suggest the possible ecological relationships between each type of bacterium and plant X. (2 marks)
Bacterium A: Bacterium B:

c) In a subsequent experiment, the effect of temperature on the growth of bacteria A and B was tested and the results are shown in the graph below:



- i) Indicate the optimum temperature for the growth of bacteria A and B respectively. (1 mark)
Bacterium A: Bacterium B:
- ii) Normal soil temperature in Hong Kong is around 30°C in summer. Global warming has led to an increase in soil temperature within 2°C. With reference to the above graph, predict the changes in the population sizes of bacteria A and B in soil in summer. (2 marks)
- d) Plant X is a foreign species that is more competitive than the native plant species in Hong Kong. With reference to your answers in (a) and (c)(ii), suggest a possible impact of global warming on the native plant community. Explain your answer. (2 marks)

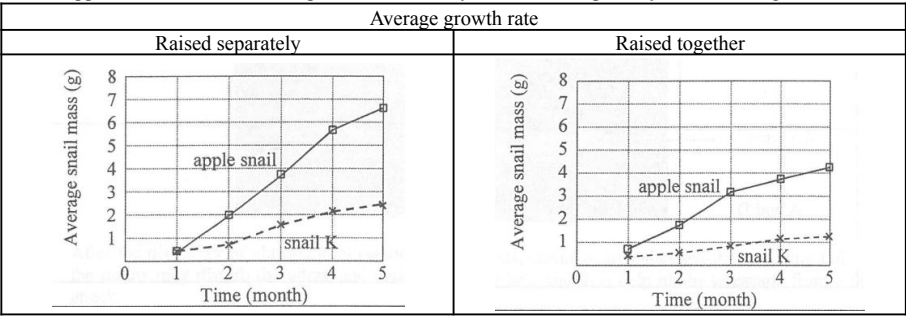
8. 2018Q11 - Essay
Sharon was told that she might be breathing out carbon dioxide containing carbon which was once a part of the body of a dinosaur, buried underground for millions of years. Use your biological knowledge to describe the journey of the carbon. The journey should include how it is possible for the carbon derived from a dinosaur buried underground to go through millions of years and appear in the form of carbon dioxide in Sharon's breath. (11 marks)

9. 2019Q7
The table below shows the changes in soil nitrogen content and the number of species of herbaceous plants and woody plants before and after a landslide on a hillside:

	Soil nitrogen content (mg g ⁻¹)	Number of plant species	
		Herbaceous plants (e.g. grass)	Woody plants (e.g. shrubs and trees)
Before landslide	6	10	15
2 years after landslide	1	17	2
20 years after landslide	3	14	9

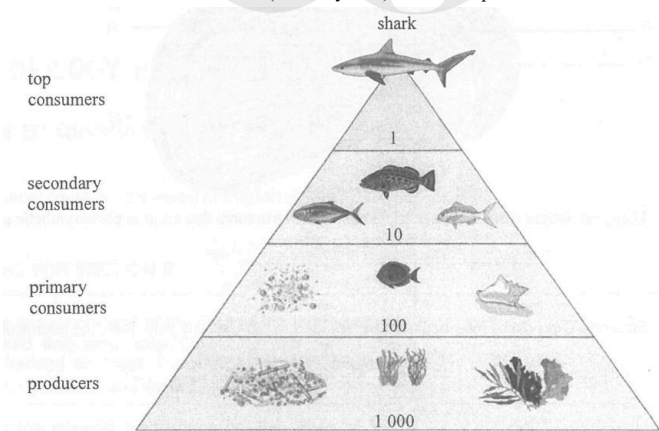
- a) What type of ecological succession occurred on the hillside after the landslide? Explain your answer. (2 marks)
- b) i) Explain the change in the soil nitrogen content shown in the above table. (3 marks)
ii) With reference to the change in soil nitrogen content, explain the change in the plant composition after the landslide in terms of the number of species of herbaceous and woody plants. (3 marks)

10. 2019Q9
The apple snail originates from South Africa. It was first imported to Asian countries for human consumption. However, it escaped to the local wetland habitat. Below are data regarding the average growth rates of apple snail and a local snail species K when they were raised separately and raised together.



- a) With reference to the above data, what would happen to the population of snail K in the wetland habitat once the apple snails have escaped to this habitat? Support your answer with data from the graphs above. (4 marks)
- b) It has been noted that apple snails consume wetland plants at a high rate, especially buds and young leaves. Suggest why the feeding habits of apple snails may have an adverse effect on the community of local wetland habitats. (3 marks)
- c) Apart from the above, suggest another biotic factor which may explain why an imported species would turn into an invasive or dominant species. (1 mark)
- d) Suggest **one** human activity which might lead to an invasion of imported species in Hong Kong. (1 mark)

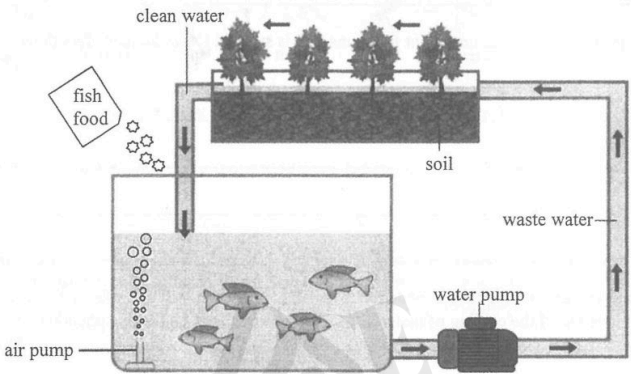
11. 2020Q2
The diagram below shows the total biomass (arbitrary unit) at each trophic level in a marine ecosystem:



- a) As the trophic level becomes higher, the total biomass of each level decreases. Give **two** reasons for this phenomenon. (2 marks)
- b) Sharks, being the top consumers in the ocean, play an important role in keeping the populations of other marine organisms under control. It is predicted that the extinction of sharks would result in the overpopulation of producers. Explain how this would happen. (3 marks)

12. 2020Q6

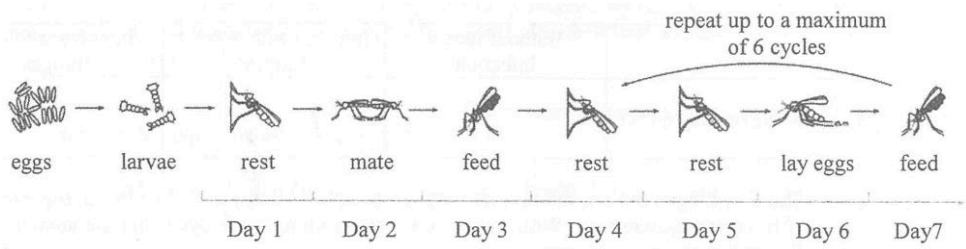
The diagram below shows a mini-ecosystem in which waste water from a fish tank is used as a source of nutrients for plant growth by making use of the interaction among fish, microorganisms and plants. Adding fertilizers and periodic change of water are not necessary. This is an eco-friendly way to grow vegetables and raise fish for human consumption.



- a) Ammonia, a toxic substance, is the major waste product excreted by fish. Ammonia in waste water from the fish tank can be converted to nitrate, which is required by plants for growth.
 - i) Name the bacteria involved in the conversion. (1 mark)
 - ii) Describe how plants can obtain nitrate from waste water and make use of it for protein synthesis in their leaves. (3 marks)
- b) The air pump performs some important functions in this system. Describe these functions (2 marks)
 - (1) _____
 - (2) _____
- c) If double the amount of fish food is added accidentally, some fish will die a few days later. (4 marks)
Based on your knowledge of the cycling of materials, suggest an explanation for this phenomenon.

13. 2020Q7b

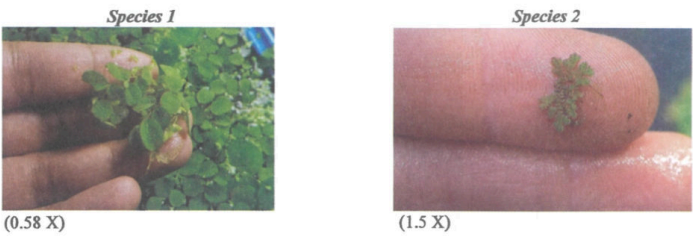
The diagram below shows the life cycle of mosquitoes:



- b) The Food and Environmental Hygiene Department introduced a fungal species to control the mosquito population in Hong Kong. Mosquitoes will die in a few days if they are infected with the fungus. What is the ecological relationship between mosquitoes and this fungal species? (1 mark)

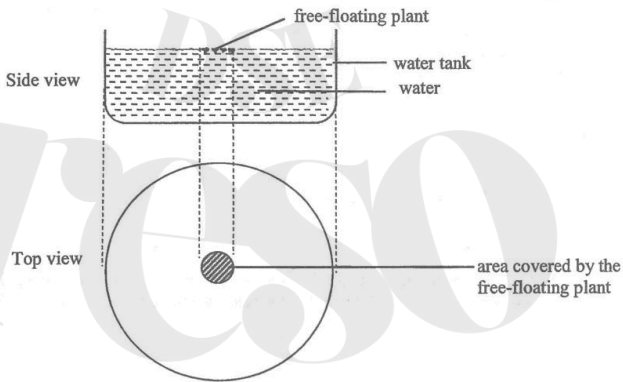
14. 2021Q9a, b

The photographs below show the appearances of two species of free-floating, freshwater plants, Species 1 and Species 2:

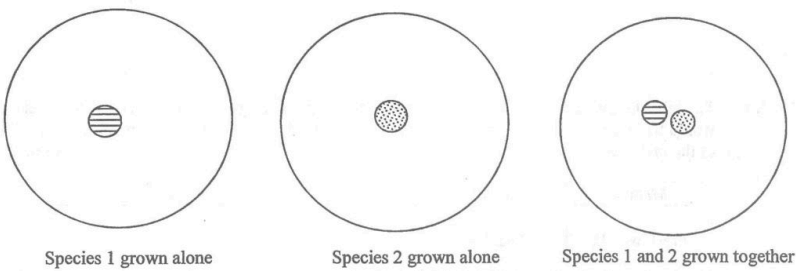


To study the interaction between these two plant species, each species was grown either alone or together with another species in a water tank for 50 days. Each species covered 10% of the area of water surface at the beginning of the experiment. The experimental set-up and design are shown in the following diagrams.

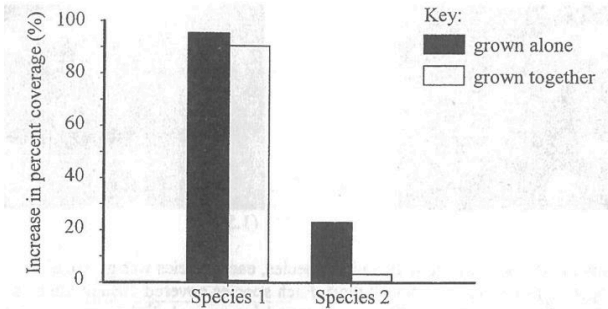
Experimental set-up:



Experimental design (top view):



The percent coverage of each plant species was measured at the beginning and at the end of the experiment. The increase in the percent coverage are shown below:

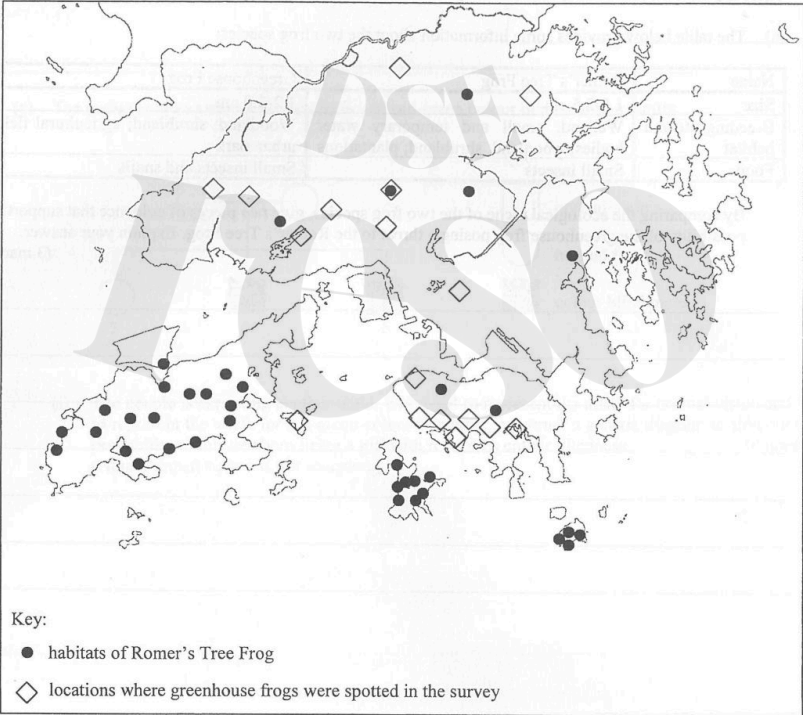


- a) With reference to the aim of the experiment, what conclusions can be drawn about the interaction between Species 1 and 2? Explain your answer.
Conclusion 1: Conclusion 2: (4 marks)
- b) With reference to the photographs of Species 1 and 2, suggest an explanation for the difference in the percent coverage of the two plant species when they were grown together. (2 marks)

15. 2023Q7
Greenhouse frog is a foreign species which is now found in many local areas according to a recent survey. There is a concern that these greenhouse frogs might threaten a local endangered species, Romer’s Tree Frog.
a) The table below provides some information about the two frog species:

Name	Romer’s Tree Frog	Greenhouse Frog
Size	1.5-2.5 cm	1.2-3.0 cm
Breeding site and habitat	Wetland, small and temporary water bodies; woodland; shrubland; plantations	Woodland; shrubland; agricultural field; urban park
Food	Small insects	Small insects and snails

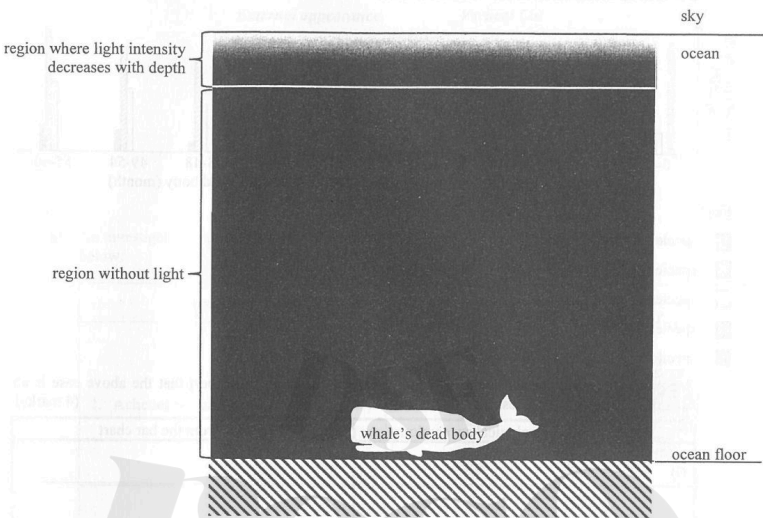
- By comparing the ecological niche of the two frog species, give **two** pieces of evidence that support the possibility of the greenhouse frog posing a threat to the Romer’s Tree Frog. Explain your answer. (3 marks)
- b) The map below shows the distribution of the two frog species in Hong Kong:



- Suggest why the information above **cannot** prove that the Romer’s Tree Frog is facing a real threat from the greenhouse frogs. (1 mark)
- c) Suggest how you could collect data to show if Romer’s Tree Frogs are facing a real threat from greenhouse frogs. (2 marks)

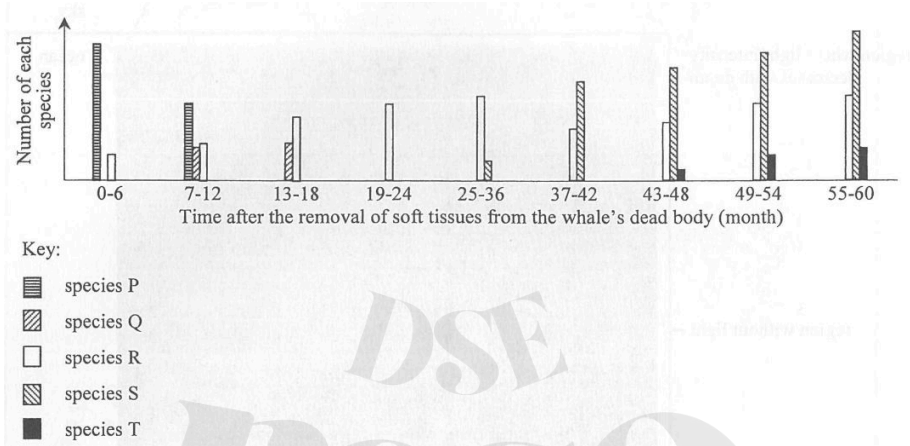
16. 2023Q8

When whales die, their dead bodies sink to the bottom of the ocean. The whale carcasses support a unique community known as whale fall community. The diagram below shows different regions of the ocean and the location of a whale's dead body:



- a) i) With reference to the energy flow in the ecosystem, what is the ultimate source of the energy stored inside the whale's dead body? (1 mark)
- ii) With reference to the above diagram, explain the importance of the whale's dead body to the whale fall community on the ocean floor. (2 marks)
- b) What is the role of the organisms that feed on the soft tissues of the whale's dead body in the cycling of materials? (1 mark)

- c) After the soft tissues of the whale's dead body have been consumed, another group of organisms start to feed on the remaining nutrients from the skeleton. For an average-sized whale, it could have 2000 - 3000 kg lipid stored inside its skeleton. The bar chart below shows the abundance of different species that feed on the skeleton of the whale over time:



Complete the following table with evidence from the bar chart to support that the above case is an example of ecological succession. (4 marks)

Characteristics of ecological succession	Evidence from the bar chart
(i)	
(ii)	

17. 2024Q7a

The following photograph was generated by an artificial intelligence programme using the following sentence:
'A photograph capturing Hong Kong students on a field trip to a rocky shore, studying the distribution and abundance of organisms along the shore.'



- a) The photograph does not truly reflect the requirement in the sentence because two pieces of essential equipment are missing.
- i) List the **two** pieces of essential equipment for the study. (1 mark)
- ii) How could you use the equipment listed in (i) to collect the data needed for the study? (2 marks)

18. 2024Q11 - Essay

Carbon footprint is an estimation of the total amount of greenhouse gases (including carbon dioxide and methane) generated by our actions, e.g. our choice of food. For instance, skipping meat one day per week will help to reduce the carbon footprint.

Discuss why the practice of eating a vegetarian diet rather than a mixed diet can reduce your personal carbon footprint by referring to the biological aspects of the practice. Briefly discuss **two** other personal actions that you can do to reduce your carbon footprint from other perspectives. (11 marks)

Ans

2012Q16: C (78%)	2012Q17: C (80%)	2014Q30: D (69%)	2014Q31: D (51%)
2014Q33: B (45%)	2015Q30: C (79%)	2015Q31: B (58%)	2016Q30: A (61%)
2016Q34: D (47%)	2017Q12: B (71%)	2017Q32: D (56%)	2017Q33: C (62%)
2017Q34: B (62%)	2017Q35: A (56%)	2018Q5: D (37%)	2018Q6: D (89%)
2018Q14: A (91%)	2018Q32: C (64%)	2018Q33: B (77%)	2019Q29: A (68%)
2019Q33: B (70%)	2019Q34: C (81%)	2020Q32: B (92%)	2021Q32: A (54%)
2021Q33: B (80%)	2021Q34: C (61%)	2021Q35: A (47%)	2021Q36: C (54%)
2022Q8: D (62%)	2022Q9: C (73%)	2022Q10: D (80%)	2023Q10: B (82%)
2023Q21: A (72%)	2023Q22: D (49%)	2023Q23: B (53%)	2024Q14: D (66%)
2024Q32: B (58%)	2024Q33: D (50%)	2024Q34: B (44%)	

2012Q6

- 6 (a) $(10\,000\,000 - 1\,000\,000) / 10\,000\,000 \times 100\% = 90\%$
Method 1 mark (1)
Correct answer 1 mark (1)
- (b) • some energy is not obtained by the organisms of higher trophic level as not the whole body of the prey is consumed / part of the food is not digestible or is egested (1)
• some energy is lost by the organisms of the higher trophic level through excretion / respiration / in the form of heat (1) (1)
- (c) • the percentage decrease in energy content is greater (1)
• as caterpillars mainly feed on leaves of trees, leaving most of the part of the biomass of trees unconsumed (1) (1)
-
- 6 marks

2013Q5

- 5 (a) nitrification (1) (1)
- (b) • nitrifying bacteria (1)
• which convert nitrite to nitrate (1) and help lower the nitrite level (2)
- (c) • adding aquatic plants to the aquarium (1)
• the plants will assimilate / absorb / convert nitrate into their biomass (1) (2)
-
- 5 marks

2013Q8

- 8 (a) • correct title (1)
- concept: distribution of species, defined location, e.g.
 - The distribution and abundance of animal species A and B on a rocky shore
 - Abundance of species A and B from the back of a shore to the waterfront
- (5)
- correct choice of axes (1)
- axes with labels and units (1)
- X: distance from the back of the shore (m); Y: abundance (number of individuals m^{-2} or m^{-2})
- correct plotting and joining of line showing the distribution of species A and B (1,1)
- (b) • species A is found to be more abundant at the back of the shore than species B (1)
- the back of the shore has a lower chance / shorter period of time to be covered by seawater (1)
- thus, organisms found at this region is more likely to face the problem of desiccation (1)
- therefore, it can be deduced that species A has a higher tolerance of desiccation (1)
- (c) • place a transect line from the back of the shore to the waterfront (1)
- place a quadrat along the transect line at regular intervals (1)
- count the number of species A and B in the quadrat (1) and record the results

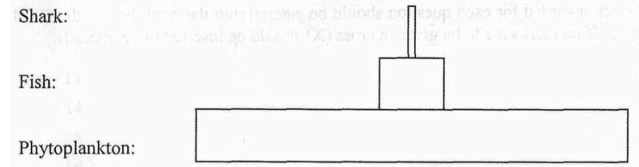
12 marks

2014Q5

- 5 (a) • secondary succession (1)
- because it involves the restoration of the community after a major disturbance, i.e. fire (1)
- (b) (i) • from year 0 to 3, herbs are the dominant vegetation (1)
- from year 3 to 8, woody plants are the dominant vegetation (1)
- (ii) • underground vegetative organs and seeds were not damaged in the fire (1),
- herbs generally germinate and grow faster than woody plants (1)
- hence the percentage cover of herbs increased rapidly at first (1) and become the dominant vegetation in the first three years
- as woody plants continue to grow and over-shadow the herbs / out-compete the herbs for light (1), they replace herbs as the dominant vegetation

8 marks

2015Q3

- 3 (a) • phytoplankton → fish → shark (1)
- (b) 
- Shark: (1)
- Fish: (1)
- Phytoplankton: (1)
- correct shape (1)
- labels (1)
- (c) • when energy in the lower trophic level is transferred to the next higher level, there is energy lost (1)
- and the individuals at lower levels are smaller in size (1)
- therefore, a larger number of individuals at a lower trophic level is required to support those at upper levels (1)
- (d) Any **two** of the following:
- by dissecting the gut of the predators to find out what prey items are inside (1)
 - field observation of the feeding relationship (1)
 - laboratory study by offering different preys to a predator (1)

8 marks

2016Q5

- 5 (a) • species B (1)
- the lower shore has a higher algal biomass which indicates that it is a more favourable area (1)
- a stronger competitor normally occupies a more favourable habitat (1)
- (b) • mean temperature at the lower shore is lower than that at the middle shore / vice versa (1)
- if temperature tolerance is a determining factor, middle shore should be unfavourable to species A / lower shore should be more favourable to species A (1)
- however, species A occupied middle shore instead of lower shore (1)
- therefore, there are other factors determining the distribution / temperature tolerance is not the determining factor (1)
- (c) • quadrat sampling relies on counting organisms in a confined area (1)
- so it is only useful for studying plants or very slow moving organisms / crabs may move away from the quadrat (1)

9 marks

2017Q8

- 8 (a) Bacterium A: inhibits the growth of plant X (1)
 Bacterium B: promotes the growth of plant X (1) (3)
 Bacterium C: has no obvious effect on the growth of plant X (1)
- (b) Bacterium A: parasitic (1)
 Bacterium B: mutualistic (1) (2)
- (b) (i) Bacterium A: $30(^{\circ}\text{C}) \pm 1$
 Bacterium B: $35(^{\circ}\text{C}) \pm 1$ (1)
- (ii) • population size of bacterium A decreases (1)
 • population size of bacterium B increases (1) (2)
- (d) • inhibition on the growth of plant X decreases as the population size of bacterium A decreases / the growth of plant X increases as the population size of bacterium B increases (1) (2)
 • hence, native plant community decreases in size / diversity due to the increased threat of plant X (1)
-
- 10 marks

2018Q11

- 11 **(A) How the carbon atom was released from dinosaur to the atmosphere (max 1 mark)**
 the dead body of the dinosaur might be
 • decomposed by microorganisms to form carbon dioxide containing the carbon atom (1)
 • turned into a portion of fossil fuels, which is then released as carbon dioxide to the atmosphere by combustion (1) max 1
- (B) The cycling of the carbon atom before reaching Sharon (max 3 marks)**
 • the carbon dioxide containing the carbon atom was used by plants in photosynthesis (1)
 • and converted to organic matter (1)
 • the carbon passed on to consumers by feeding the plants / passed along the food chain (1)
 • the carbon atom in the biomass of these organisms would be released to the atmosphere by respiration / decomposition (1) max 3
 this cycle repeated until the carbon atom finally reached Sharon
- (C) How Sharon obtained the carbon atom (max 2 marks)**
 • Sharon obtained the carbon atom by feeding / from the food chain (1)
 • the food containing the carbon atom was digested and absorbed in the small intestine (1)
 • Sharon breathes in the carbon dioxide containing the carbon atom and breathes out (1) max 2
- (D) What happened inside Sharon's body (max 2 marks)**
 • the absorbed food containing the carbon atom travels along the circulatory system (1)
 • and is taken up by body cells for respiration (1)
 • to form carbon dioxide (1)
 • the carbon dioxide containing the carbon atom travels along the circulatory system to the lungs for gas exchange / breathing out (1) max 2
- (E) Effective communication (0-3)** max 3
-
- 11 marks

2019Q7

- 7 (a) • secondary succession (1) (2)
 • because succession occurred in pre-existing soil / pre-existing plants (1)
- (b) (i) • after the landslide, the original vegetation / top soil was removed, so the soil nitrogen content fell / after the landslide, rapid growth of herbaceous plants absorbed nitrogen from the soil, so the soil nitrogen content fell (1)
 • some plant species have symbiotic nitrogen fixing bacteria in their root nodules, which enrich the soil nitrogen content by fixing atmospheric nitrogen (1) (max. 3)
 • as time went by, the dead mass of the new vegetation was decomposed by soil bacteria (1)
 • hence releasing nitrogen compounds back into the soil (1), and so soil nitrogen content increased with time
- (ii) • in the early stage / 2 years after the landslide, there were many more herbaceous plant species than woody plant species (1)
 • since the nutrient requirements of herbaceous plant species are lower than those of woody plant species (1), low soil nitrogen content favours the growth of herbaceous plants / does not favour the growth of woody plants (3)
 • subsequently, the number of woody plants species increased as the soil nitrogen content continued to increase (1)
-
- 8 marks

2019Q9

- 9 (a) • both apple snails and snail K showed a lower growth rate when they were raised together (1)
 • this shows that there was competition between apple snails and snail K for habitat and / or food (1)
 • the growth rate of snail K would be slower and some of them would fail to reach maturity for reproduction / death rate of snail K would be higher (1)
 • so the population of snail K would fall / decline (1) (4)
 OR
 • apple snails had a higher growth rate when raised separately from snail K (1),
 • the growth rate of snail K fell significantly when raised together with apple snails (1)
 • this shows that apple snails were more competitive than snail K (1)
 • hence the population of snail K would fall / decline (1)
- (b) • in the long term, plants would fail to reach maturity for reproduction (1)
 • it reduce the size of the plant community in the habitat (1),
 • this would reduce food availability / shelter for other animals (1) (3)
 OR
 • in the short term, plants would grow more slowly (1)
 • the biomass of producers would decrease (1)
 • this would reduce food availability for other herbivores / animals (1)
- (c) • lack of natural predators in the local habitat / a high rate of reproduction (1) (1)
- (d) any reasonable answers such as:
 • release of imported pets to the natural environment once owners cannot / do not want to keep them anymore (1)
 • release of imported organisms to the natural environment for religious purposes (1) (1)
 • bringing in non-local seeds / plants which then reproduce and spread to the wild habitat through seed dispersal (1)
-
- 9 marks

2020Q2

- 2 (a) biomass of organisms at lower trophic levels is not 100% available to organisms at higher trophic levels because: (any **two** of the following)
- not all body parts are consumed (1) (2)
 - indigestible parts are egested (1)
 - energy loss at each trophic level through respiration / excretion / heat (1)
 - if sharks become extinct, the population of the secondary consumers will not be kept under control and thus will increase (1)
 - these secondary consumers will prey on the primary consumers, leading to a great drop in the population of the primary consumers (1) (3)
 - as a result, the producers will become too numerous because the number of primary consumers has decreased (1)

5 marks

2020Q6

- 6 (a) (i) nitrifying bacteria (1) (1)
- (ii) • nitrates are absorbed by roots through diffusion and active transport (1) and transported to the xylem
- by transpiration pull, nitrates are transported upward along the xylem vessels (1) to the leaf cells (3)
- in the leaf cells, nitrates combine with intermediate products from photosynthesis to form amino acids, which are used in protein synthesis (1)
- (b) Any **two** of the following:
- to ensure there is sufficient oxygen for the respiration of organisms in the system (fish / plant roots / nitrifying bacteria) (1)
 - to provide oxygen for decomposition / degradation / breaking down of organic waste in fish excreta into inorganic waste (1)
 - to provide oxygen for nitrification (1) (2)
- (c) • the ammonia produced by fish increases as the amount of food is doubled (1)
- the bacteria fail to convert all ammonia to nitrates (1)
- as a result, the ammonia will accumulate in the water (1)
- to a certain level which is lethal to the fish / will kill the fish (1)
- OR
- the fish cannot consume all of the food / excess food is left in the water / organic waste produced by fish increases as the amount of food is doubled (1) (4)
 - putrefying bacteria / decomposers decompose the excess food / organic waste to form ammonia (1)
 - as a result, the ammonia will accumulate in the water (1)
 - to a certain level which is lethal to the fish / will kill the fish (1)

10 marks

2020Q7b

- 7 (b) parasitism (1) (1)

2021Q9a, b

- 9 (a) • they were each other's competitor (1) because their percent coverage / growth rate when grown together is lower than when grown alone (1) (4)
- Species 1 was more competitive than Species 2 (1) as the drop in coverage / growth rate for Species 2 was much higher than that for Species 1 (1)
- (b) • larger leaves enabled Species 1 to be more successful in overtopping / shading Species 2 (1) (2)
- and so increased its competitiveness for light / surface space (1)

2023Q7

- 7 (a) Concept for mark award
- identify overlapping niches from the information as evidence (1x2) (3)
 - concept of competition resulting from similar ecological niches (1)

e.g.

Any **two** of the following as evidence:

- both feed on small insects (1)
- they have a similar body size (1)
- there is some degree of overlap in their habitats (1)
- these show that their ecological niches are similar / overlapping, i.e. there may have been keen competition between them (1) therefore the greenhouse frog may threaten the Romer's Tree Frog

- (b) Any **one** of the following:
- greenhouse frogs have yet to be found in the natural range of Romer's Tree Frog / the distribution of the two frogs has very little overlap now (1) (1)
 - the abundance of the two frogs are not known (1)
- (c) • record the number of greenhouse frogs and Romer's Tree Frogs in areas where their habitats overlap over a period of time / put the two frogs in a simulated habitat and record their number over a period of time (1) (2)
- if the number of greenhouse frogs increases while Romer's Tree Frog decreases (1) it may imply that the existence of the greenhouse frog has posed a threat to Romer's Tree Frog

6 marks

2023Q8

- 8 (a) (i) • sunlight / light energy from the sun / the sun (1)
- (ii) • no sunlight can reach the sea bottom, so there are no photosynthetic producers down there (1)
- therefore the dead bodies of the whales become an important source of energy / food for the organisms at the bottom of the sea (1) (2)
- (b) • they decompose the organic matter to inorganic matter for the cycling of materials / they are decomposers (1) (1)
- (c) Concept for mark award:
• quote relevant data with explanation of how it supports the example as ecological succession (1+1) x2 (4)

e.g. (any **two** sets:)

Characteristics of ecological succession	Evidence from the bar chart
dominant species is replaced over time	(any two dominant species with time reference) e.g.: - species P is the dominant species in month 0-12, followed by species R in month 13-36 - species R is the dominant species in month 13-16, and is then replaced by species S in month 37-60+ - species P is the dominant species in the early stage, then is replaced by species R, and species S becomes dominant in the later stage
composition / structure of the community changes with time	(a description showing the change in composition) e.g. species P and R appear first in month 0-6, later change to species P, Q and R in month 7-12
climax community	species R, S and T remain stable from month 49 to 60

8 marks

2024Q7a

	Concept for mark award	Example	Marks
7(a)(i)	point out the missing equipment	line transect and quadrat	(1)
7(a)(ii)	how to use the missing equipment	lay a transect from backshore to waterfront and place a quadrat next to the transect	(1)
		count the number of species and their abundance within the quadrat	(1)

2024Q11

	Concept for mark award	Example	Marks
11	why vegetarian diet can reduce carbon footprint:		max. 6
	<u>food chain</u> comparison of the length of food chains	food chain involved in a vegetarian diet is shorter than that in a diet with meat / eating vegetables involves fewer trophic levels in the food chain than eating meat	(1)
	carbon footprint of the additional trophic level	animals in the additional trophic level release carbon dioxide through respiration / decomposition of waste / excretory products / faeces / uneaten parts from each trophic release carbon dioxide	(1)
	inference	therefore, carbon dioxide released from animals / carbon footprint from meat will not be counted if we adopt a vegetarian diet	(1)
	relating methane emissions to plant-eating mammals in the mixed diet	in addition, plant-eating mammals such as cows emit a large amount of methane during digestion of plant materials, which contributes to the carbon footprint of a mixed diet	(1)
	relating the practice to change in land use	more people adopting a vegetarian diet would reduce the demand for meat, so land use for rearing plant-eating mammals or cows will decrease, and the carbon footprint from methane emissions from rearing plant-eating mammals will also be reduced	(1)
	<u>gas exchange</u> crops undergo photosynthesis, which absorbs carbon dioxide	during cultivation of crops, they undergo photosynthesis during daytime, which absorbs carbon dioxide and releases oxygen	(1)
	growing crops reduces carbon dioxide from atmosphere	therefore adopting a vegetarian diet would lead to a reduction in carbon dioxide in the atmosphere	(1)
	relating the practice to change in land use	more people adopting a vegetarian diet would increase the demand for crops, and more land used for growing crops means that more carbon dioxide would be removed from the atmosphere	(1)

	Concept for mark award	Example	Marks
	other personal actions (should be from other perspectives)	any two of the following points: • take public transport instead of private transport to reduce carbon dioxide release through transportation • buy local products instead of imported products to reduce the carbon dioxide from transportation • use electrical appliances with high energy efficiency to save electricity, which reduces emission of carbon dioxide from combustion of fossil fuels • use energy saving lighting such as LED to reduce energy use, which reduces emission of carbon dioxide from combustion of fossil fuels • turn off electrical appliances when not in use to reduce electricity use, which reduces emission of carbon dioxide from combustion of fossil fuels	max. 2 (1) (1) (1) (1) (1)
	effective communication		(0-3)

11 marks